

eFuse Controller

This chapter describes the eFuse controller.

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33.1 Overview

Electrically programmable fuses (eFuses) are used to configure the device after deassertion of $\overline{\text{PORRST}}$. The eFuse values are read and loaded into internal registers as part of the power-on-reset sequence. The eFuse values are protected with single bit error correction, double bit error detection (SECDED) codes. These fuses are programmed during the initial factory test of the device. The eFuse controller is designed so that the state of the eFuses cannot be changed once the device is packaged.

33.2 Introduction

The eFuse controller automatically reads the values of the eFuses and shifts them into registers during the power-on reset sequence. No action is required from the application code. However, in a safety critical application, the user code should check to see if a correctable or an uncorrectable error was detected during the reset sequence and then preform a self-test on the eFuse controller ECC logic.

33.3 eFuse Controller Testing

33.3.1 eFuse Controller Connections to ESM

There are three connections from the eFuse controller to the Error Signaling Module (ESM). If an uncorrectable error occurs during the loading of the eFuse values after reset, a group three, channel one error and a group one channel 40 error are sent to the ESM. The group three error will cause the $\overline{\text{ERROR}}$ pin to go low. If during the eFuse loading a correctable error occurs, only a group one channel 40 error is sent to the ESM. If an error occurs during the eFuse controller self test, then a group one channel 41 error and a group one channel 40 error are sent to the ESM. After reset, by default, the group one errors do not affect the $\overline{\text{ERROR}}$ pin. If the software enables the appropriate bit in the appropriate ESM Influence Error Pin Set/Status Register (ESMIEPSRn) while the group one error is set, the $\overline{\text{ERROR}}$ pin will go low.

Table 33-1. ESM Signals Set by eFuse Controller

ESM Signal	Uncorrected Load Failure	Correctable Load Error	Self Test		
			eFuse Self Test	eFuse stuck at 0 Test	
				Version a: with Error pin	Version b: without Error pin
Group 3 Channel 1	X			X	
Group 1 Channel 40	X	X	X		
Group 1 Channel 41			X	X	X

33.3.2 Checking for eFuse Errors After Power Up

For safety critical systems, it is required that you check the status of the eFuse controller after a device reset. A suggested flow chart for checking the eFuse controller after device reset is shown in [Figure 33-1](#). Failures during the eFuse self test can be grouped into three levels of severity. Depending on the safety critical application, the error handling for each error type may be different.

33.3.2.1 Class 1 Error

A class 1 error of the eFuse controller means that there was a failure during the autoloading sequence. The values read from the eFuses cannot be relied on. All device operation is suspect. A class 1 error is indicated by a signal to group 3 channel 1 of the ESM. This will cause the $\overline{\text{ERROR}}$ pin to go active low.

33.3.2.2 Class 2 Errors

A class 2 error is an indication that the safety checks of the eFuse controller did not work. These are also serious errors because you can no longer guarantee that a more severe error did not occur.

33.3.2.3 Class 3 Error

A class 3 error indicates that there was a single bit failure reading the eFuses that was corrected by ECC bits. Proper operation is still likely, but the system is now at a higher risk for a future non-correctable error. When a correctable error occurs, ESM group 1, channel 40 will be set. In the suggested flow chart shown in [Figure 33-1](#) below, the single bit error is determined by directly reading the eFuse error status register, and not depending on the integrity of the connections between the eFuse controller and the ESM.

33.3.2.4 Stuck at Zero Test

The purpose of the stuck at zero test is to verify that the eFuse controller could signal the ESM if an autoload error did occur. It basically verifies the path through the eFuse controller and to the ESM. This is done by writing a special instruction to the eFuse controller boundary register, then verifying that the proper bits are set in the eFuse controller pins register. Upon successful completion of this test ESM group 1 channel 41 and ESM group 3 channel 1 will be set. This will force the **ERROR** pin low.

- Version A
 - Write boundary register (address 0xFFFF8C01C) with 0x003FC000 to set the error signals.
 - Read pins register (address 0xFFFF8C02C) and verify that bits 14, 12, 11 and 10 are set.
 - Write boundary register (address 0xFFFF8C01C) with 0x003C0000, to clear the error signals.
 - Verify that ESM group 1 channel 41 and group 3 channel 1 are set, then clear them.

If the system cannot support a test which causes the **ERROR** pin to go low, then the stuck at zero test can be modified as follows:

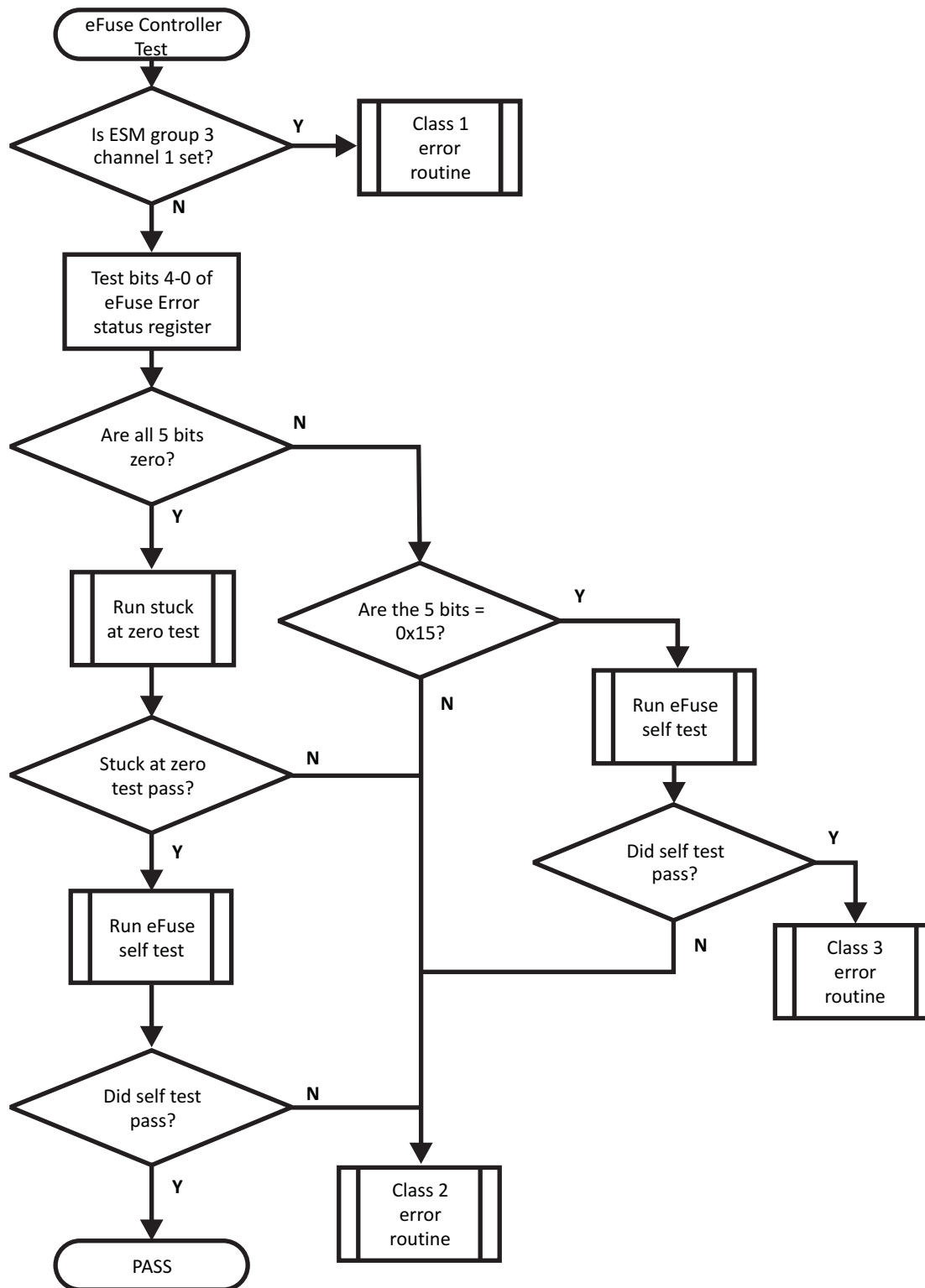
- Version B
 - Write boundary register (address 0xFFFF8C01C) with 0x003BC000.
 - Read pins register (address 0xFFFF8C02C) and verify that bits 14, 12, and 11 are set.
 - Write boundary register (address 0xFFFF8C01C) with 0x003C0000, to clear the error signals.
 - Verify that ESM group 1 channel 41 is set, then clear it.

This alternate method provides less test coverage because the path from the uncorrectable error signal from the eFuse controller to the ESM is not specifically tested. However, even if this path is broken, reading the five eFuse error status bits will indicate that an error occurred.

33.3.2.5 eFuse ECC Logic Self Test

The eFuse controller self test performs extensive validation of the ECC logic in the eFuse controller. This test should only be performed once for every device **PORRST** cycle. Perform the self test by following these steps:

- Write 0x00000258 to the self test cycles register (EFCSTCY) at address 0xFFFF8C048.
- Write 0x5362F97F to the self test signature register (EFCSTSIG) at address 0xFFFF8C04C.
- Write 0x0000200F to the boundary register at address 0xFFFF8C01C. This triggers the self test. The test takes 610 VCLK cycles to complete. The application can poll bit 15 of the pins register at address 0xFFFF8C02C to wait for the test to complete.
- Check ESM group 1 channels 40 and 41 for any errors, neither should be set.
- Verify that bits 4 to 0 of the eFuse Error Status register at address 0xFFFF8C03C are zero.

Figure 33-1. eFuse Self Test Flow Chart


33.4 eFuse Controller Registers

All registers in the eFuse Controller module are 32-bit, word-aligned; 8-bit, 16-bit and 32-bit accesses are allowed. [Table 33-2](#) provides a quick reference to each of these registers. Specific bit descriptions are discussed in the following subsections. The base address for the control registers is FFF8 C000h.

Table 33-2. eFuse Controller Registers

Offset	Acronym	Register Description	Section
1Ch	EFCBOUND	EFC Boundary Control Register	Section 33.4.1
2Ch	EFCPINS	EFC Pins Register	Section 33.4.2
3Ch	EFCERRSTAT	EFC Error Status Register	Section 33.4.3
48h	EFCSTCY	EFC Self Test Cycles Register	Section 33.4.4
4Ch	EFCSTSIG	EFC Self Test Signature Register	Section 33.4.5

33.4.1 EFC Boundary Control Register (EFCBOUND)

[Figure 33-2](#) and [Table 33-3](#) describe the EFCBOUND register. The eFuse Boundary Control Register is used to test the connections between the eFuse controller and the ESM module. The eFuse Boundary Control Register is also used to initiate an eFuse controller ECC self-test.

Figure 33-2. EFC Boundary Control Register (EFCBOUND) [offset = 1Ch]

31								24									
Reserved																	
R-0																	
23		22		21		20		19		18		17		16			
Reserved				EFC Self Test Error		EFC Single Bit Error		EFC Instruction Error		EFC Autoload Error		Self Test Error OE		Single Bit Error OE			
R-0				R/W-0		R/W-0		R/W-0		R/W-0		R/W-0		R/W-0			
15		14		13		12								8			
Instruction Error OE		Autoload Error OE		EFC ECC Selftest Enable		Reserved											
R/W-0		R/W-0		R/W-0		R-0											
7						4		3								0	
Reserved										Input Enable							
R-0										R/W-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after power-on reset (nPORRST)

Table 33-3. EFC Boundary Register (EFCBOUND) Field Descriptions

Bit	Field	Value	Description
31-22	Reserved	0	Read returns 0. Writes have no effect.
21	EFC Self Test Error	0 1	This bit drives the self test error signal when bit 17 (Self Test Error OE) is high. This signal is attached to ESM error Group 1, Channel 41. Drives the self test error signal low, if Self Test OE is high. Drives the self test error signal high, if Self Test OE is high.
20	EFC Single Bit Error	0 1	This bit drives the single bit error signal when bit 16 (Single bit Error OE) is high. This signal is attached to ESM error Group 1, Channel 40. Drives the self test error signal low, if Single Bit Error OE is high. Drives the self test error signal high, if Single Bit Error OE is high.

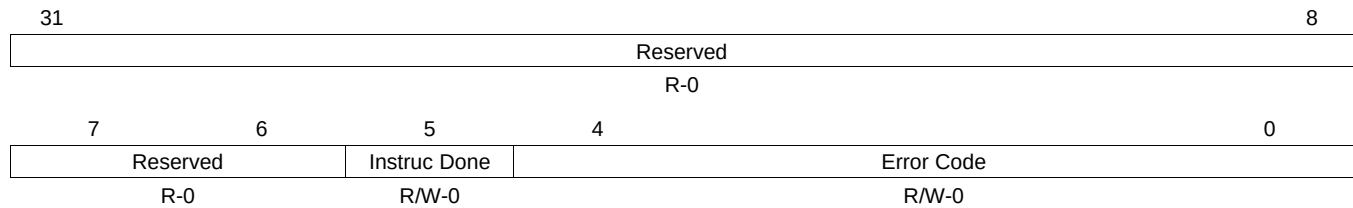
Table 33-3. EFC Boundary Register (EFCBOUND) Field Descriptions (continued)

Bit	Field	Value	Description
19	EFC Instruction Error	0 1	This bit drives the instruction error signal when bit 15 (Instruction Error OE) is high. This signal is used to denote an error occurred during e-fuse programming. This signal is not attached to the ESM. Drives the Instruction Error signal low, if Instruction Error OE is high. Drives the Instruction Error signal high, if Instruction Error OE is high.
18	EFC Autoload Error	0 1	This bit drives the Autoload Error signal when bit 14 (Autoload Error OE) is high. This signal is attached to ESM error Group 3, Channel 1. Drives the Autoload Error signal low, if Autoload Error OE is high. Drives the Autoload Error signal high, if Autoload Error OE is high.
17	Self Test Error OE	0 1	The Self Test Error Output Enable bit determines if the EFC Self Test signal comes from the eFuse controller or from bit 21 of the boundary register. EFC Self Test Error comes from eFuse controller. EFC Self Test Error comes from the boundary register.
16	Single Bit Error OE	0 1	The single bit error output enable signal determines if the EFC Single Bit Error signal comes from the eFuse controller or from bit 20 of the boundary register. EFC Single Bit Error comes from eFuse controller. EFC Single Bit Error comes from the boundary register.
15	Instruction Error OE	0 1	The instruction error output enable signal determines if the EFC Instruction Error signal comes from the eFuse controller or from bit 19 of the boundary register. EFC Instruction Error comes from eFuse controller. EFC Instruction Error comes from the boundary register.
14	Autoload Error OE	0 1	The autoload error output enable signal determines if the EFC Autoload Error signal comes from the eFuse controller or from bit 18 of the boundary register. EFC Autoload Error comes from eFuse controller. EFC Autoload Error comes from the boundary register.
13	EFC ECC Selftest Enable	0 1	The eFuse Controller ECC Selftest Enable bit starts the selftest of the ECC logic if the four input enable bits (EFCBOUND[3:0]) are all 1s. No action Start ECC selftest if EFCBOUND[3:0] are Fh.
12-4	Reserved	0	Read returns 0. Writes have no effect.
3-0	Input Enable	Fh All others	The eFuse Controller ECC Selftest Enable bit starts the selftest of the ECC logic if the four input enable bits (EFCBOUND[3:0]) are all 1s. ECC selftest can be started if EFC ECC Selftest Enable, bit 13, is set ECC selftest cannot be started.

33.4.3 EFC Error Status Register (EFCERRSTAT)

Figure 33-4 and Table 33-5 describe the EFCERRSTAT register.

Figure 33-4. EFC Error Status Register (EFCERRSTAT) [offset = 3Ch]



LEGEND: R/W = Read/Write; R = Read only; -n = value after power-on reset (nPORRST)

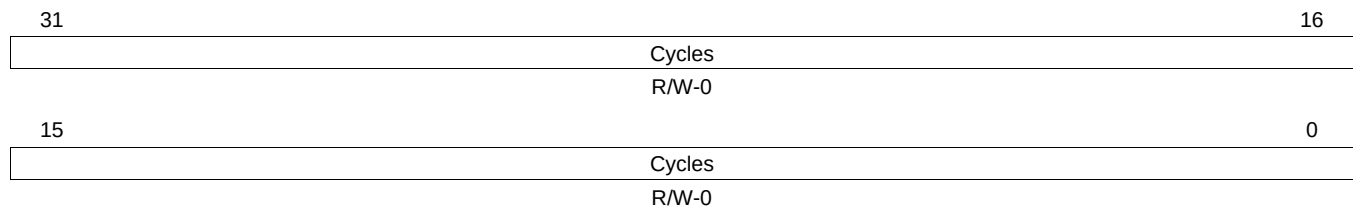
Table 33-5. EFC Error Status Register (EFCERRSTAT) Field Descriptions

Bit	Name	Value	Description
31–6	Reserved	0	Reads return zeros, writes have no effect.
5	Instruc Done	0 1	Instruction done. Used to indicate that the eFuse self test has completed The eFuse controller is still executing. The eFuse controller has completed executing.
4–0	Error Code	0 5h 15h 18h All other values	The error status of the last instruction executed by the eFuse Controller No error. An uncorrectable (multibit) error was detected during the power-on autoloading sequence. At least one single bit error was detected and corrected during the power-on autoloading sequence. The signature generated by the ECC self-test logic did not match the golden signature written in the EFCSTSIG register. The EDAC circuitry might have a fault. All other values are reserved for e-fuse system tests and are not expected to occur in normal system use.

33.4.4 EFC Self Test Cycles Register (EFCSTCY)

Figure 33-5 and Table 33-6 describe the EFCSTCY register.

Figure 33-5. EFC Self Test Cycles Register (EFCSTCY) [offset = 48h]



LEGEND: R/W = Read/Write; -n = value after power-on reset (nPORRST)

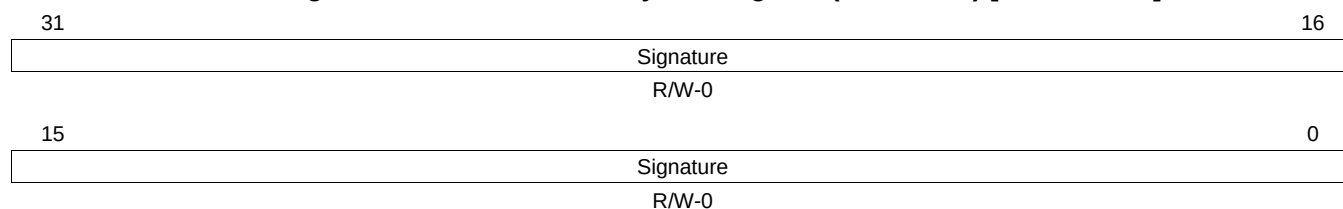
Table 33-6. EFC Self Test Cycles Register (EFCSTCY) Field Descriptions

Bit	Name	Description
31–0	Cycles	This register is used to determine the number of cycles to run the eFuse controller ECC logic self test. It is recommended to use a value of 600 (0x00000258).

33.4.5 EFC Self Test Signature Register (EFCSTSIG)

Figure 33-6 and Table 33-7 describe the EFCSTSIG register.

Figure 33-6. EFC Self Test Cycles Register (EFCSTSIG) [offset = 4Ch]



LEGEND: R/W = Read/Write; -n = value after power-on reset (nPORRST)

Table 33-7. EFC Self Test Cycles Register (EFCSTSIG) Field Descriptions

Bit	Name	Description
31–0	Signature	This register is used to hold the expected signature for the eFuse ECC logic self test. It is recommended to write a value of 0x5362F97F to this register and a value of 600 (0x00000258) to the EFCSTCY register. If after running the eFuse ECC logic self test, the calculated signature does not match the expected signature in the EFCSTSIG register, then a value of 18h is stored in the EFCERRSTAT register.